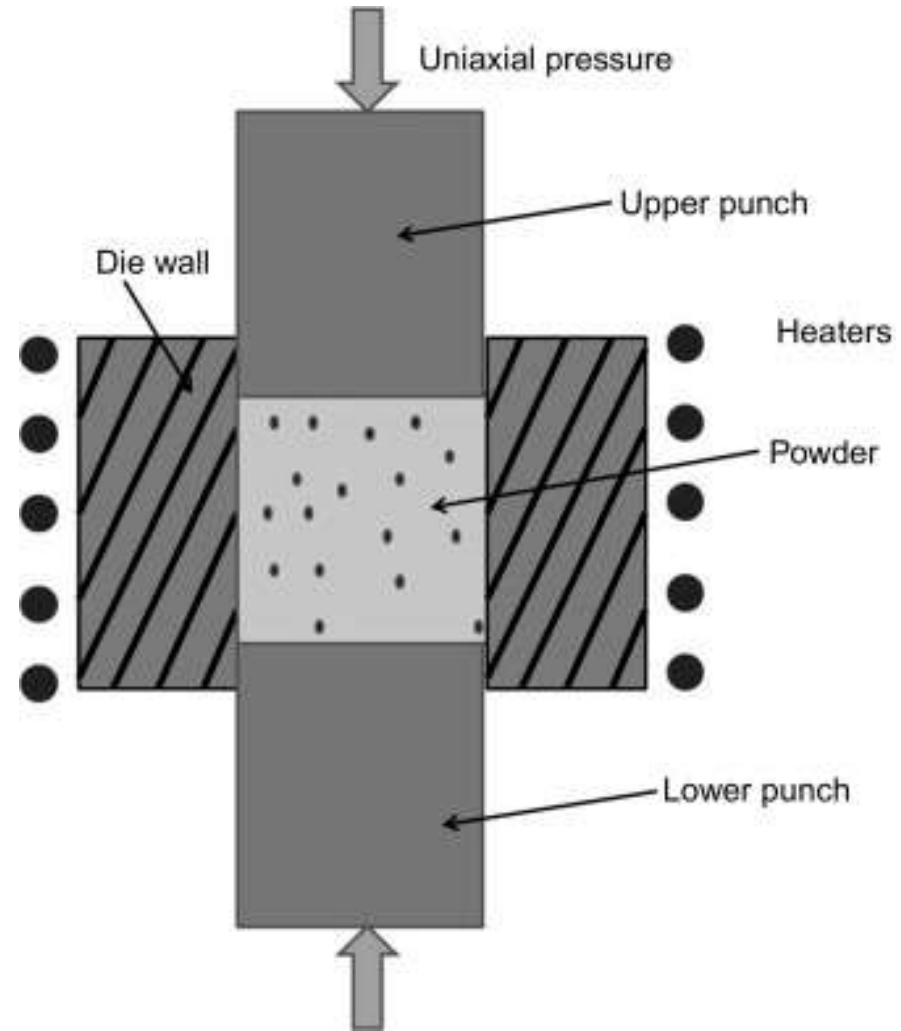


- High-temperature compaction of metal powders refers to the process in which pressure is applied to the powder at elevated temperatures, thus, **combining the consolidation and sintering operations into a single step.**
- Fully dense parts of metal powder with precisely controlled microstructure can be produced by hot consolidation.
- This can be carried out by hot pressing, hot extrusion, hot rolling, Hot Isostatic Pressing (HIP) as well as by hot forging of preforms.
- In high-temperature compaction, the powder must be protected from reaction with air and hence pressing in dies must be carried out under protective atmosphere or by '**canning**' the powder, that is, encapsulating the powder in a container to prevent reaction with atmosphere.
- This makes the process complex and an expensive one.

PRINCIPLES

- When a powder aggregate is subjected to high pressure and temperature, densification occurs in **three stages**.
- In the **first stage**, the aggregate is almost instantaneously compacted to high density by **plastic deformation of the powder particles**. When the external pressure reaches equilibrium with the internally developed stresses, no further closure of pores is possible by plastic deformation. However, with sufficiently high pressures, densities as high as 95% can be attained during this stage.
- In the **second stage** of densification, various **creep mechanisms** become operative. But with increasing pore closure, the stresses become low and diffusional creep becomes more prominent.
- In the **third stage**, with the increased curvature of pore surfaces, **diffusion becomes the predominant densification mechanism** and is similar to normal sintering.

Uniaxial Hot Pressing



- The technique of hot pressing of powder is very versatile and widely used for powders that are difficult to sinter.
- In hot pressing, the pressure is generally applied during the heating cycle, reaching a maximum during the interval at the maximum temperature.
- In case of certain materials, **pressure must be maintained during part of the cooling period also for best results.**
- Parts produced in this way have excellent physical properties, high density and usually a minimum of preferred orientation, though the technique is an expensive one.
- However, proper scheduling and acceleration of heating and cooling cycles can justify the use of hot pressing in special cases where excellent properties are needed or where a limited number of special parts are required.

- One of the principal problems in hot pressing of metal powders is the choice of a suitable die material.
- It must be strong enough at the hot pressing temperature to withstand the applied pressure without plastic deformation.
- It also should not react with the powders being hot-pressed.
- Protective atmospheres are necessary to prevent oxidation of powders and die material.
- Hot pressing tools are made of molybdenum, tungsten, stellite, die steels, cemented carbides, or graphite. Alumina dies as well as titanium boride are also used as die materials.
- Metallic dies and punches are water cooled to avoid plastic deformation at high temperatures.
- For very high temperatures, graphite, cemented carbide or superalloy dies are commonly employed.
- Graphite is cheap, readily available, easy to machine and has a low-expansion coefficient. Graphite dies are unique in that they are useful at temperature as high as 2,200 to 2,500°C, but die life is greatly reduced at temperatures above 500°C, if oxidizing conditions prevail.
- If proper measures are taken to inhibit carbon contamination, they may be used with a wide variety of materials.
- Use of vacuum can help not only in doing away with inert gases, but also improves better reactivity of powders during pressing through degassing.

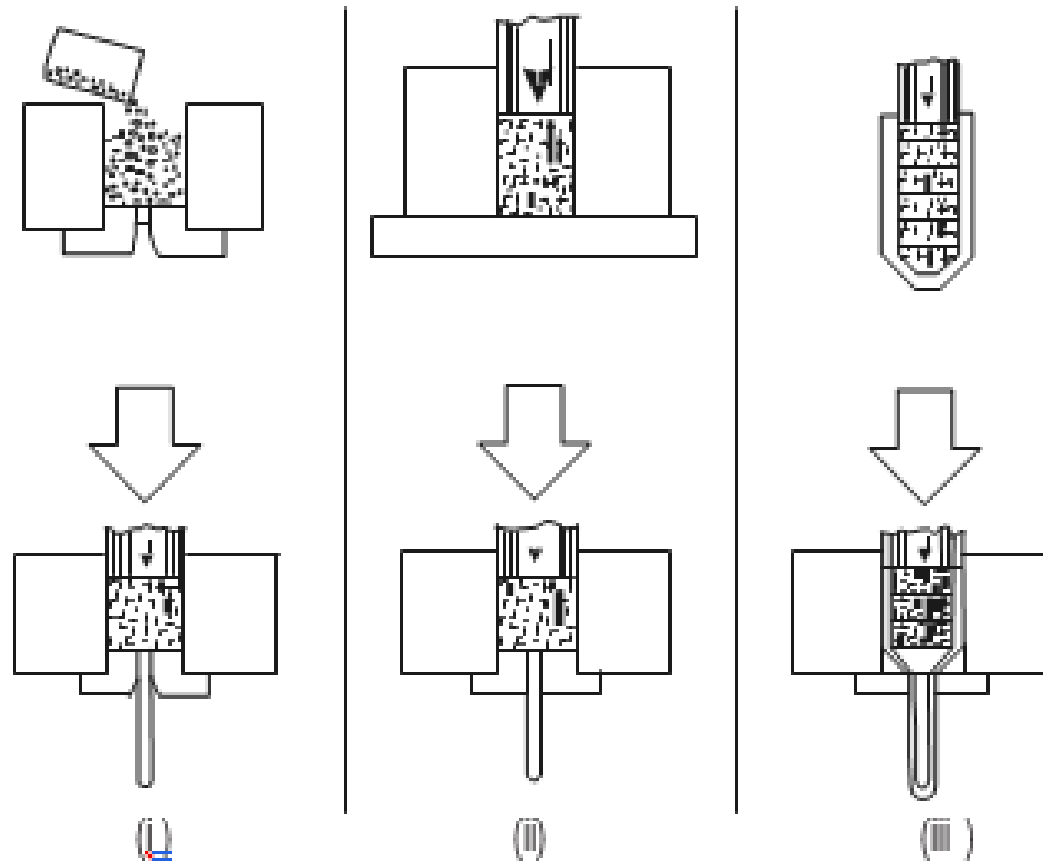
Hot Extrusion

- Powder extrusion differs from conventional extrusion only in the need to have a canned preform prior to extrusion and removing the canning material after extrusion.
- Hot compaction and hot mechanical working are combined in hot extrusion to produce fully dense products.
- Hot extrusion uses high temperature, high pressure as well as mechanical deformation to densify and shape the metal powder into standard mill forms or tubular forms.
- During extrusion, hydrostatic compressive stresses and shear forces act on the powders.
- In powder extrusion, the powder mixture is precompacted into a billet by pressing and sintering, hot pressing or by hot die pressing.

- The can serves as a container not only for degassing, removal of moisture, annealing of powders before extrusion but also to protect the powder from contamination from the atmosphere, extrusion lubricants and tooling.
- The can material is selected on the basis of its hot plasticity, its reactivity, ease of removal and cost.
- Mild steel is a widely used canning material, though copper is also used.
- The extrusion route has been adapted for consolidating Sintered Alumina Powder (SAP) as well as structural parts from aluminium powders having remarkable properties.
- It has also been successfully applied to several cermet compositions by using such base metals as lead, aluminium and copper.
- Powder extrusion is also widely used in the

Powder extrusion routes:

(i) extrusion of loose powders, (ii) cold pressing and extrusion, and (iii) canning followed by extrusion



- In the **first method**, loose powder is placed in a heated container and extruded directly through a die.
- No atmospheric protection is provided, and the heat of the container is enough to raise the temperature of the powder for extrusion.
- **This process has been used to extrude magnesium alloy powders.**
- In the **second method**, the powder is cold compacted, hot-pressed and subsequently extruded.
- This process has been successfully used to produce **aluminium alloy billets and molybdenum powder compacts.**
- In the **third route**, the powders are 'canned', that is, placed in a metallic capsule and sealed after degassing by evacuation at room or elevated temperature.
- The canned powder is heated and extruded along with the can.

Spark Sintering

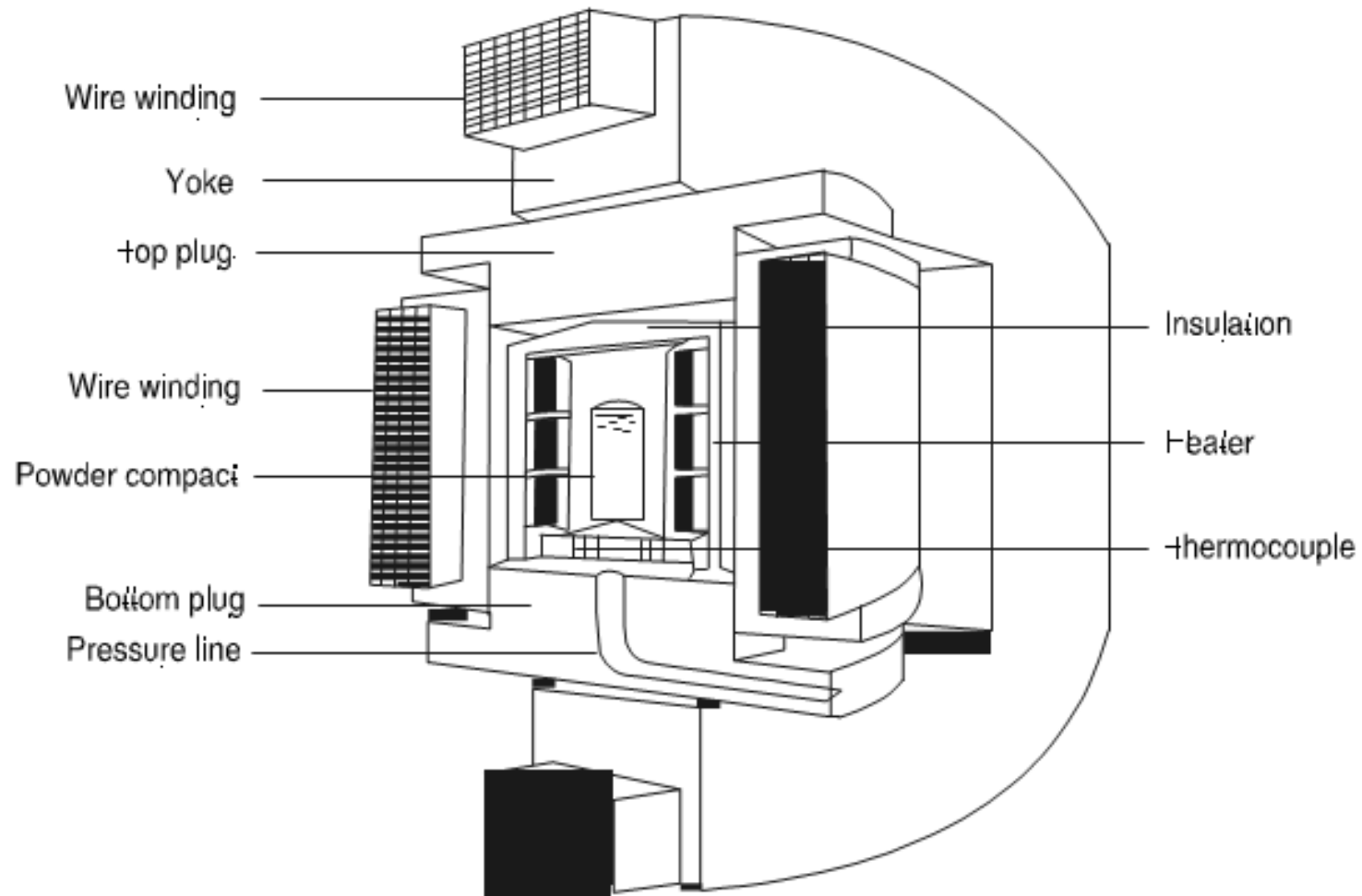
- In spark sintering, the powders or a green powder preform is placed between two punches, which also serve as **electrodes** for conducting a low voltage, high amperage current through the powder.
- The powder is heated to the sintering temperature by the applied current and is simultaneously pressed.
- The chief advantage of this process is that the **die remains relatively cold during** the operation.
- Resistance sintering of a mixture of atomized

Hot Isostatic Pressing (HIP)

- Developed simultaneously in the US and Sweden about 50 years back.
- This technique was first used to densify nuclear reactor parts as well as sintered carbide tools.
- Today, it has emerged as an ideal choice for consolidation of powders of nickel and cobalt-base superalloys, tool steels, maraging steels, titanium alloys, refractory metal powders, ceramic powders as well as cermets.
- Also be used for fabricating components of aluminium, magnesium, copper-base alloys, cemented tungsten carbides, magnetic materials as well as metal matrix composites.
- Also used for a variety of other applications like bonding of dissimilar material, consolidation of plasma coatings, processing soft and hard magnetic

- HIP is the application of pressure at elevated temperatures to obtain net or near-net shape parts from metal, ceramic and cermet powders.
- The process offers increased design flexibility and as the process is not shape or size dependent, it results in optimised material usage.
- HIP results in enhanced quality and improved mechanical properties.
- Tooling is simple and scrap is minimised with little or no machining required.

Cross-section of HIP unit

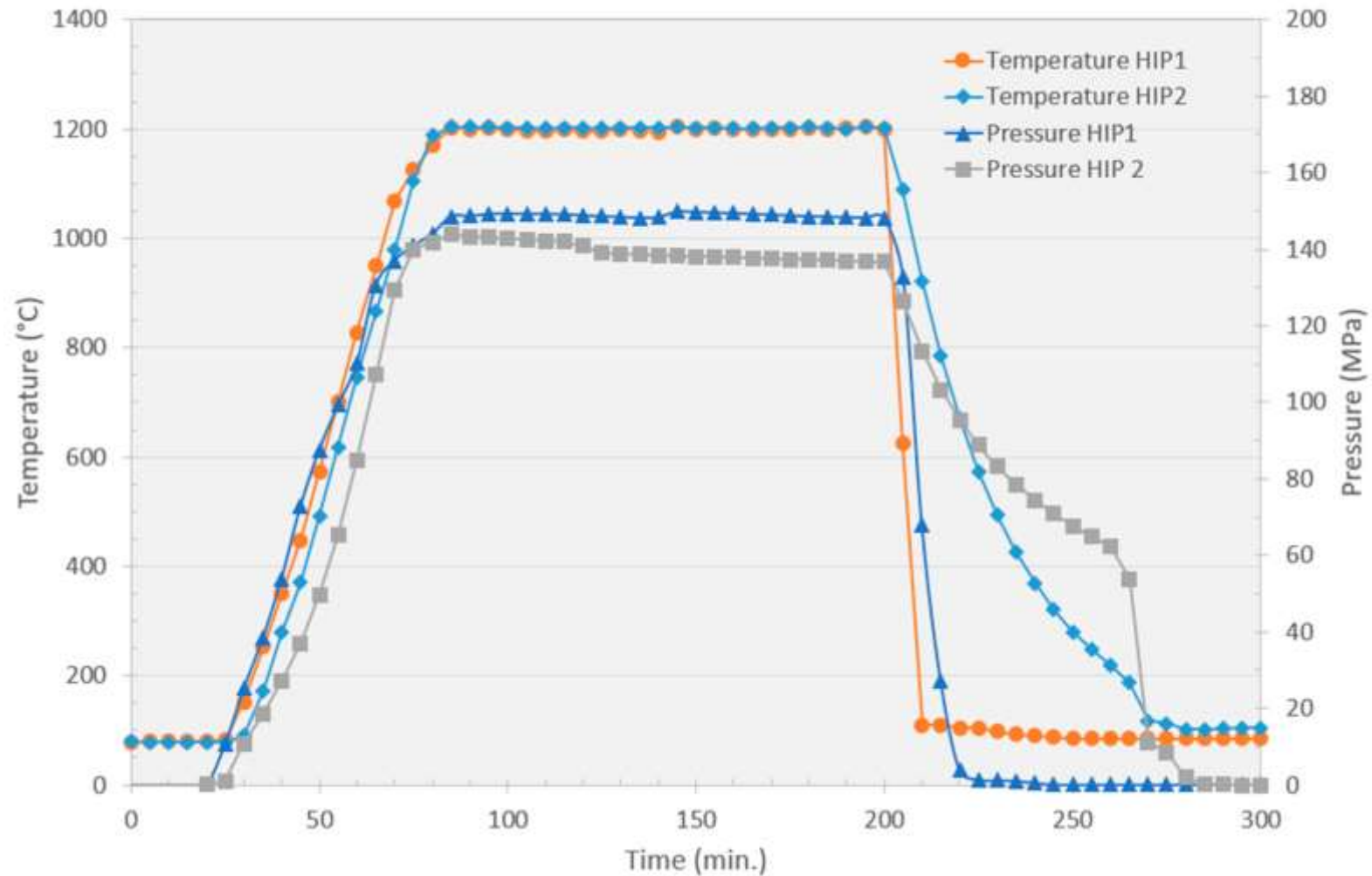


Major components of a HIP unit

- Pressure vessel
- High temperature furnace
- Pressurising system
- A water-cooled pressure vessel within which a furnace, thermally insulated from the pressure vessel is kept.
- The pressure vessel is usually made of low alloy steel.
- The basic function of the unit is to heat the powders while applying uniform gas pressure on

- Furnaces used are of radiation or convection type heating furnaces with graphite or molybdenum heating elements, though nichrome is also used.
- The furnace heats the material to be pressed, while a pressurising medium, usually a gas, is used to apply a high pressure during the process for specific times.
- The normal pressurising medium used in HIP is argon, though nitrogen, helium or even air can be used.
- These gases must be of very high purity (less than 5 ppm total impurities).
- The design of the equipment is very complex and

A typical HIP cycle



- Elastomer are not used because of high temperature.
- Generally, a metal container (called a "**can**") or glass is used as a flexible, leak-proof mould in Hot Isostatic Pressing.
- This mould is degassed after filling with powder to remove volatile components and then sealed.
- The pressurising gas, usually argon, is then let into the vessel and then a compressor is used to increase the pressure to the desired level.
- The furnace is then started and both temperature and pressure are raised to the required values.
- HIP presses are available in diameters up to 2 m with pressures ranges from 40 to 300 MPa and temperatures ranging from 500 to 2200°C, though

- The processing time can last up to 4 hours depending on the material and the size of the part.
- The furnace is then allowed to cool, followed by depressurising the chamber and removal of parts.
- The process results in full density parts with isotropic properties, even in large and complex shaped parts.
- During HIP, the pores present not only get closed by flow of matter by diffusion and creep, but also bonded across the interface to form a continuous material.
- However, HIP, cannot remove pores present on the surface because the pressurising medium will be in contact with these pores.

- Low temperature and high isostatic pressure gives a structure with controlled grain growth and isotropic properties.
- This gives superior performance as in the case of ceramic powders and sintering of diamond tools.
- Powders characteristics such as apparent density and particle size play an important role during processing.
- Powders having a high apparent density, such as spherical powders are preferred for HIP.
- Hot isostatic pressing is used when great uniformity is required, and where net-shape production (of also intricate and complex shapes) leads to economical advantages (e.g. metal alloys for aerospace applications and complex ceramic parts).

- Hot Isostatic Pressing of large components,

- Depending upon the materials being pressed, working temperature range and pressure applied as suitable choice of **heating elements** can be made. Commonly used heating elements include:

(i) **Kanthal** heating elements for temperatures up to **1,200°C**. These elements are oxidation-resistant and permit loading and unloading of the work pieces in the hot condition.

(ii) **Molybdenum heating elements** for temperatures in the range of **1,200 to 1,700°C**. These elements are used mainly for densification of materials prone to surface contamination.

(iii) **Graphite heating elements** for temperatures in the range of **2,000°C to 2,600°C**.

Allow pressing of materials in **either an argon or nitrogen atmosphere**.

- Usually heating elements are arranged in multiple banks, separated vertically, to obtain temperature uniformity and each bank can be controlled independently.
- The gas can be recycled for re-use.
- A close circuit cooling water system is used to control the temperature of the pressure vessel wall and covers.

Typical Temperature Ranges and Pressures

<i>Material</i>	<i>HIP temperature range (°C)</i>	<i>Applied pressure (MPa)</i>
Al alloy	350–500	100
Cu alloy	500–900	100
High-speed steel	1,000–1,200	100
Ni base superalloy	1,170–1,280	100–150
Ti alloy (Ti-6Al-4V)	880–960	100
Al ₂ O ₃	1,350–1,450	100
Ba Ti O ₃	100–1,200	100
Partially stabilized zirconia	1,350–1,500	100
SiAlON	1,700–1,800	100
WC-Co	1,300–1,350	30–100

- In a recent development, **both sintering and Hot Isostatic Pressing** of parts have been combined and done using the HIP unit.
- In this process, pre sintered component or powders (encapsulated in a sealed container) is placed inside the high-pressure vessel.
- The vessel is then evacuated, raised to sintering temperature and at a predetermined stage, high-pressure argon is introduced and pressed directly into a near-net shape part.
- This process, also known, as **Pressure Assisted Sintering (PAS)** is a possible replacement for vacuum sintering followed by HIP in a separate apparatus.
- Sintered parts having a relative density of 92% or more without any gas porosity or surface porosity can be fully densified without canning using this technique.
- The sinter-HIP versions of the process achieves improved strength and wear resistance of some

(i) ceramic insulators
showing preform, machined
and finished parts, (ii) HSS
tool



(i)



(ii)

Major benefits of HIP

(i) Enhanced density and integrity of the parts:

This means rejections and reworking can be minimised, resulting in potential savings in time and money.

(ii) Improved properties: Mechanical properties are improved due to zero or reduced porosity.

The dynamic properties, in particular fracture toughness and also the **fatigue limit are greatly improved.**

Injection Moulding

- Injection moulding of metal powders is similar to plastic injection moulding and uses the same equipment.
- Injection moulding uses very fine particles, 10 micron or less.
- The powder to be compacted is mixed with large quantities of a thermoplastic binder.
- Uniformity of the mix and a proper coating of all powder particles with the binder are essential prerequisites for injection moulding.
- Solvent extraction and evaporation techniques are used for the de binding.
- This is followed by sintering and post sintering operations.